

5 Shell colour in the brown-lipped snail, *Cepaea nemoralis*, involves more than one gene.

Gene 1 has three alleles: C^B , C^P and C^Y . These alleles form a dominance hierarchy.

- The dominant allele, C^B , results in a brown shell.
- The allele C^P results in a pink shell.
- The recessive allele, C^Y , results in a yellow shell.
- The allele C^B is dominant to C^P and the allele C^P is dominant to C^Y .

Gene 2 has two alleles, N and n .

- The dominant allele, N , results in a shell without stripes.
- The recessive allele, n , results in a shell with stripes.

Figure 5.1 shows a snail with a pink shell without stripes.



Figure 5.1

- (a) List **all** the possible genotypes of the snail shown in Figure 5.1.

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- (b) A student predicted the shell colour of the F1 offspring that are produced when a cross was carried out between a snail with a brown shell with stripes and a snail with a yellow shell without stripes. Each parent snail was known to be homozygous for gene 1 and homozygous for gene 2.

The student wrote down the cross between the two parent snails:

parental phenotypes	brown with stripes	×	yellow without stripes
parental genotypes	$C^B C^B nn$	×	$C^Y C^Y NN$

- (i) State the genotype and phenotype of the F1 offspring of the cross between these two parent snails.

genotype

phenotype

[2]

- (ii) In a second breeding experiment, snails with the genotype $C^Y C^Y nn$ were used in a test cross carried out on the F1 offspring.

The student assumed that each F1 snail would produce four different gametes in equal numbers. The student predicted the results of the test cross between $C^Y C^Y nn$ and the F1 offspring. They predicted that the offspring would show a 1 : 1 : 1 : 1 ratio of phenotypes.

Explain how an understanding of meiosis caused the student to assume that there would be four different gametes in equal numbers.

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- (iii) The experimental results of this cross did **not** match the prediction of a 1 : 1 : 1 : 1 ratio of phenotypes. Instead, half of the test cross offspring snails were brown with stripes and half were yellow without stripes.

Suggest why the experimental results differed from the predicted 1 : 1 : 1 : 1 ratio.

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- (c) Snails of different colours and stripe patterns do not occur in the same numerical proportions in different habitats such as woodland and rough grass.

Scientists compared snail populations in woodland habitats with snail populations in rough grass habitats.

- The scientists counted the number of snails, noting the number of yellow snails and the number of snails without stripes in each population.
- The results were converted to the percentage of yellow snails and the percentage of snails without stripes in each population.
- The scientists sampled populations in many areas of woodland and many areas of rough grass.

Figure 5.2 shows these results plotted on a graph.

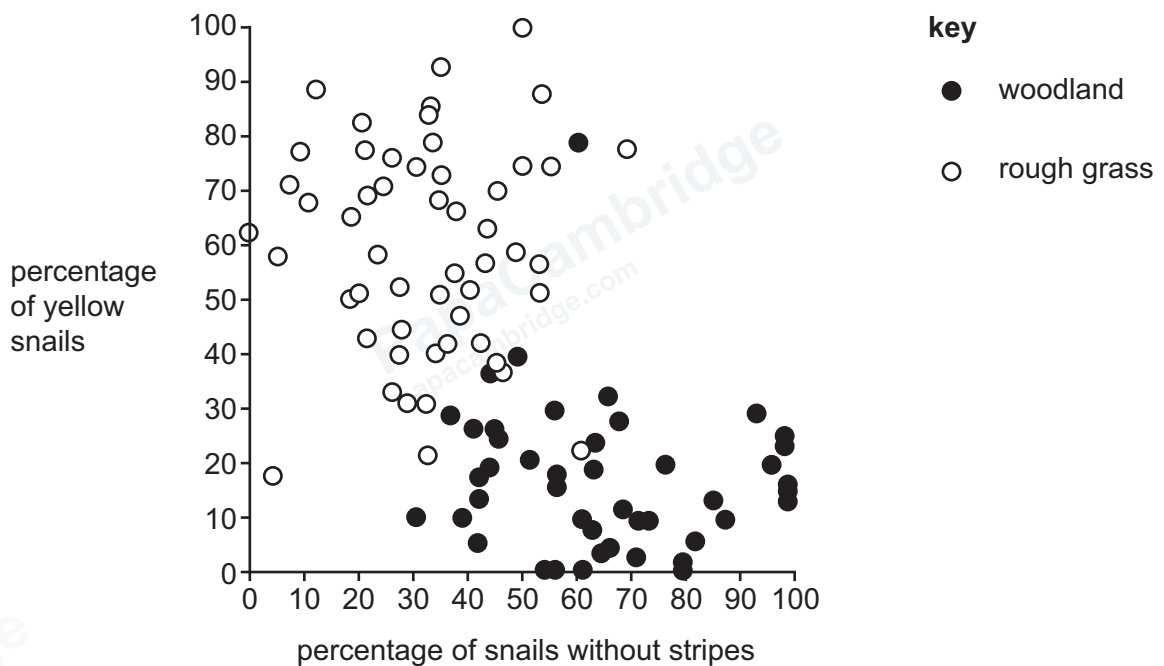


Figure 5.2



The scientists hypothesised that the selection pressure of birds eating snails was important in determining the proportions of snails with different colours and stripe patterns in woodlands and rough grass.

Explain how the selection pressure of birds eating snails might lead to the results shown in Figure 5.2.

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